📘 MODULE 11: LOOKUPS

11.1 Concept

A **lookup** enriches events with data from an external table. Your events contain src_ip=10.0.0.5, and the lookup table says 10.0.0.5 = Finance laptop, owner Alice. Splunk joins them at search time.

11.2 Why it matters

Logs hold IDs, IPs, and codes, but analysts need meaning. SOC uses:

- Asset and user context (department, criticality)
- Threat intel (known-bad IPs and domains)
- Allowlists/exception lists (service accounts)
- Code translation (4625 to “Failed logon”)

11.3 Types

Type	Notes
CSV (file-based)	Most common. Small static tables.
KV Store	Database-style, editable, supports updates. Awareness only for now.
External / geospatial	Script-based and iplocation/geo lookups. Awareness only.

11.4 The three parts of a CSV lookup

1. Lookup TABLE FILE → users.csv (the data)

2. Lookup DEFINITION → a named reference to the file (users_lookup)

3. Permissions → who can use it (must be shared, or it only works for you)

Best practice is to create a **definition** and use that name in searches.

11.5 Commands

Command	Type	Purpose
lookup	Streaming (per event)	Adds columns from the table to your events
inputlookup	Generating (needs leading |)	Reads the table as search results
outputlookup	Output	Writes search results into a lookup file

Syntax

spl

| lookup <table> <lookup_field> [AS <event_field>] OUTPUT <lookup_field> [AS <new_name>]

- <lookup_field>: the **column in the CSV** used to match.
- AS <event_field>: the **field in your events** it should be compared to. Needed when the names differ.
- OUTPUT: columns to add. It **overwrites** an existing field with the same name.
- OUTPUTNEW: adds the column **only if the event field is empty**.

11.6 Hands-on lab (do this)

Step 1: create the file. In your Ubuntu terminal:

bash

cat > /tmp/users.csv << 'EOF'

username,department,criticality

admin,IT,high

splunk-system-user,Platform,low

alice,Finance,medium

EOF

- cat > file << 'EOF' writes everything until the line EOF into the file.
- Verify with cat /tmp/users.csv. You should see 4 lines (header + 3 rows).

Step 2: upload it. Splunk Web → **Settings → Lookups → Lookup table files → New Lookup Table File**. Destination app: search. Choose /tmp/users.csv. Destination filename: users.csv. Save.

Step 3: create the definition. Settings → Lookups → Lookup definitions → New. Name: users_lookup. Type: File-based. File: users.csv. Save.

Step 4: share it. On the lookup file and definition, click **Permissions → All apps** (or at least App). Otherwise “lookup works for me but not for others/other apps” appears.

Step 5: test.

Query 1: read the table

spl

| inputlookup users_lookup

What it does: displays the CSV as a results table.
Expected result: 3 rows with username, department, criticality.
Common mistake: forgetting the leading |, since inputlookup is a generating command.

Query 2: enrich events

spl

index=_internal sourcetype=splunkd_access earliest=-24h

| stats count by user

| lookup users_lookup username AS user OUTPUT department, criticality

What it does: for each user value, finds the matching username row and adds department and criticality.
Line-by-line: stats gives one row per user · lookup users_lookup names the definition · username AS user maps CSV column to event field · OUTPUT lists the columns to add.
Expected result: admin gets IT/high. Users not in the table get blank values.
Common mistakes:

- Field name mismatch (username vs user) with no AS.
- **Case or trailing spaces** differ (Admin ≠ admin). CSV lookups are case-sensitive by default.
- Lookup not shared.

Interview explanation: "A lookup adds context to events at search time without changing the indexed data. I match a field in the event to a column in the table and output the extra columns."

Query 3: filter with the result (the SOC pattern)

spl

index=* sourcetype=linux_secure "Failed password"

| rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"

| lookup threat_ips.csv ip AS src_ip OUTPUT threat_name

| where isnotnull(threat_name)

What it does: keeps only events whose source IP appears on your threat list. (Create threat_ips.csv with columns ip,threat_name to try it.)

Query 4: write a lookup

spl

index=_internal sourcetype=splunkd_access

| stats count by clientip

| outputlookup clients_seen.csv

outputlookup creates or overwrites the file. Add append=true to add rows instead of replacing. Create a **definition** for it if you want to reference it by name.

11.7 Automatic lookups

Applies a lookup to a sourcetype **without typing lookup in every search**. UI: **Settings → Lookups → Automatic lookups → New**. Choose the sourcetype, the lookup definition, the match field, and the output fields. Under the hood it writes LOOKUP-<name> in props.conf:

ini

[linux_secure]

LOOKUP-users = users_lookup username AS user OUTPUT department criticality

Why it matters: analysts see enriched fields automatically.

11.8 Troubleshooting: "lookup not working"

Check	Fix
Run | inputlookup users_lookup	Does it return rows? If not, upload or definition problem
Permissions	Share the file and the definition with the right app
Field names	Add AS mapping, check spelling
Case or whitespace	Normalize with lower() / trim() in eval, or set case-insensitive matching in the definition's advanced options (case_sensitive_match = false)
Works manually, not automatically	Automatic lookup bound to the wrong sourcetype, or permissions/app context
New column empty	Event field was null before the lookup

11.9 Interview Q&A

- **What is a lookup?** A table used to enrich events with extra fields at search time.

- **lookup vs inputlookup?** lookup enriches existing events, inputlookup reads the table itself as the result set.
- **OUTPUT vs OUTPUTNEW?** OUTPUT overwrites existing fields, OUTPUTNEW only fills empty ones.
- **Automatic lookup?** A lookup configured in props.conf so it applies to a sourcetype automatically.
- **Lookup works for you but not a colleague?** Permissions are private.

11.10 Revision

File → definition → permissions · lookup <name> col AS eventfield OUTPUT cols · inputlookup needs | · case-sensitive by default · automatic lookup = LOOKUP- in props.conf.